

Aerospace Control Systems · Politecnico di Milano · A.Y. 2025/2026

ANT-X — Nominal and Robust Attitude Control

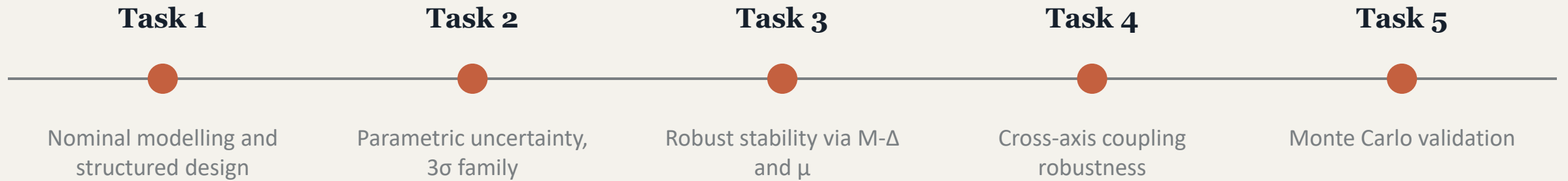
Robust multivariable control of the lateral and longitudinal dynamics of a multirotor UAV

Gozzi Federico

Leandri Simone

Graziani Matteo Marwan

How the project is organized



Objective and rationale

- **(The Objective):** The primary objective is to synthesize a reliable attitude controller for the ANT-X research quadrotor, operating on models linearized around the hovering condition.
- **(The Rationale):** To bridge theory and real-world flight, we identify grey-box models from flight tests. Our approach prioritizes a nominal design followed by strict robustness verification, explicitly quantifying every uncertainty source.

Linearised design models

Lateral axis

- State $x = [v, p, \phi]$, input δ_{lat}
- $A = [Y_v \ Y_p \ g ; 0 \ L_p \ 0 ; 0 \ 1 \ 0]$
- $B = [Y_\delta ; L_\delta ; 0]$, outputs p and ϕ
- Roll mode $L_p = +2.6478 \text{ 1/s}$

- Third row is pure kinematics: $d\phi/dt = p$
- It contributes a fixed integrator at the origin
- Second row is the isolated rate mode, driven by L_p
- First row couples gravity into the translational velocity

Longitudinal axis

- State $x = [u, q, \theta]$, input δ_{lon}
- $A = [X_u \ X_q \ -g ; 0 \ M_q \ 0 ; 0 \ 1 \ 0]$
- $B = [X_\delta ; M_\delta ; 0]$, outputs q and θ
- Pitch mode $M_q = -2.6478 \text{ 1/s}$

Observability and Controllability

Before designing the controllers: each axis' nominal model needs to be checked, we will revise if controllability and observability are satisfied:

Controllability – OK

$$\text{rank} [B \ AB \ A^2 B] = n$$

rank(C) = 3/3 on both axis. Every mode is reachable from the actuator so the closed-loop can be placed

Observability– PARTIAL

$$\text{rank} [C \ CA \ CA^2] = n$$

rank(O) = 2 the outputs [p, φ, q, θ] don't see the velocity modes v and u, the attitude and rate modes are observable

- **Detectable** ⇒ the design is well posed. The unobservable modes (v and u) are stable, so they need no feedback.

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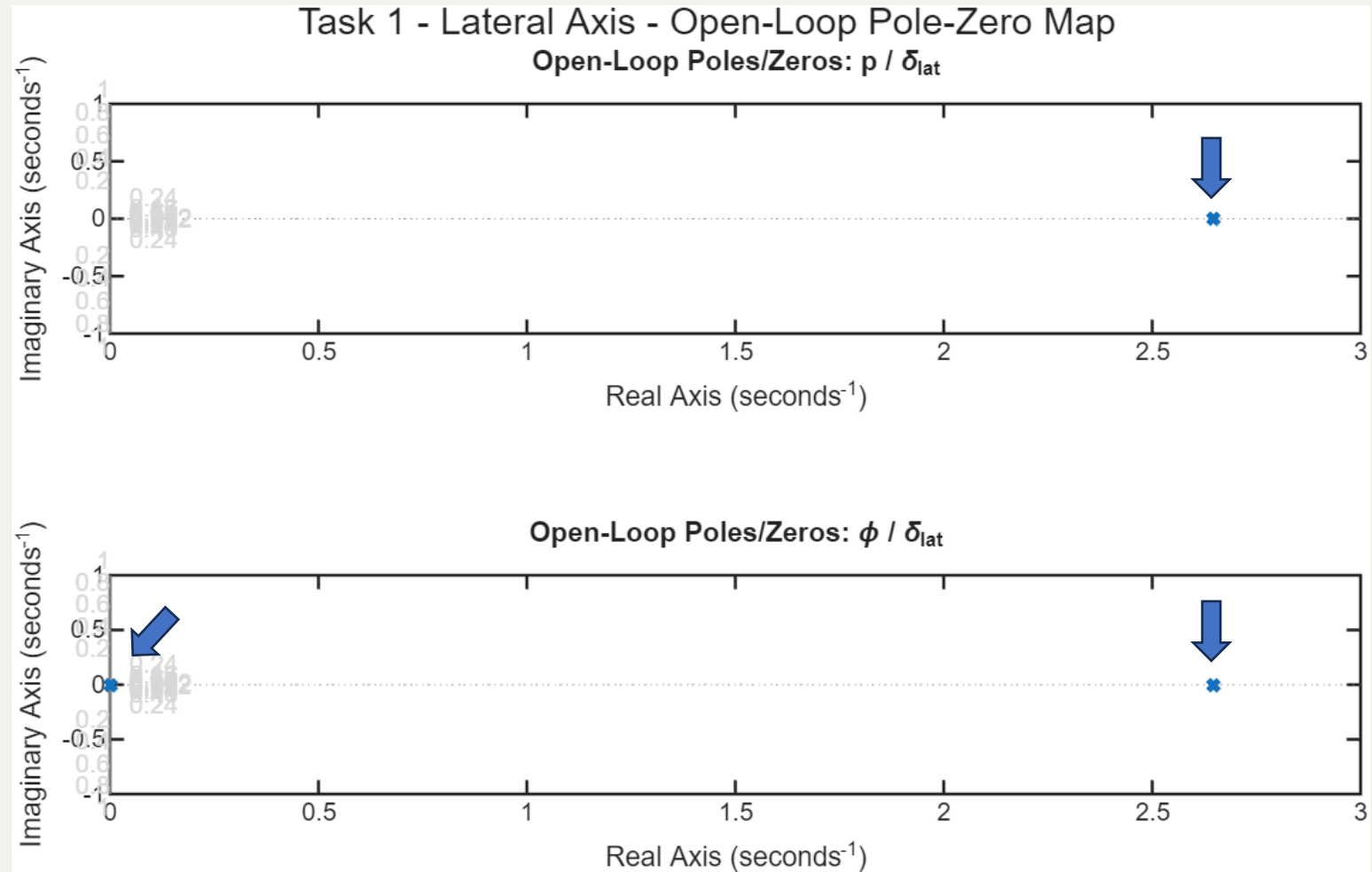
Nominal modelling and design

Task 1 — lateral and longitudinal axes

Lateral axis — poles and zeros

Lateral axis

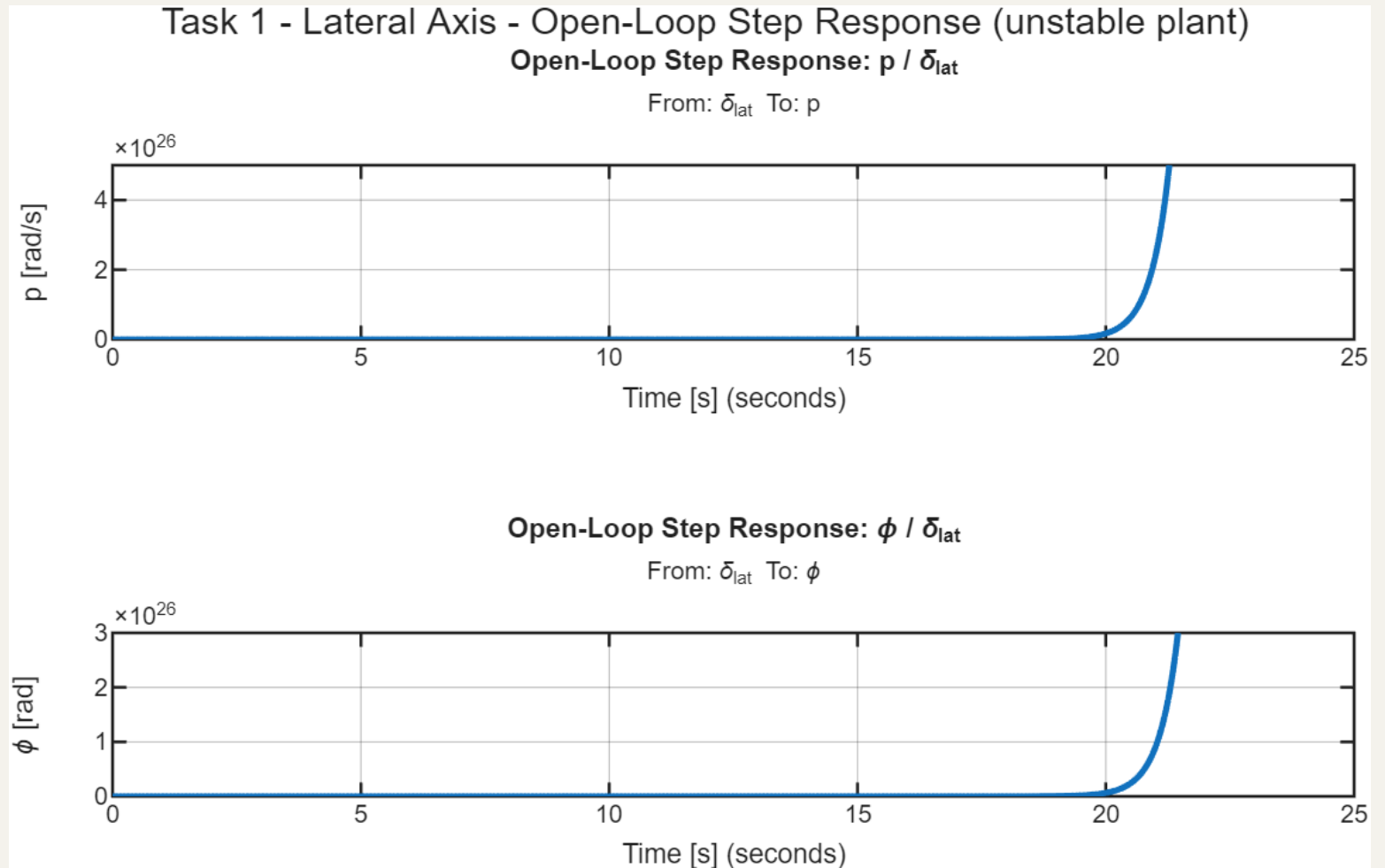
- Roll pole at $\text{Re} = +2.6478$: the plant is **unstable**
- Integrator at the origin



Lateral axis — step response

Lateral

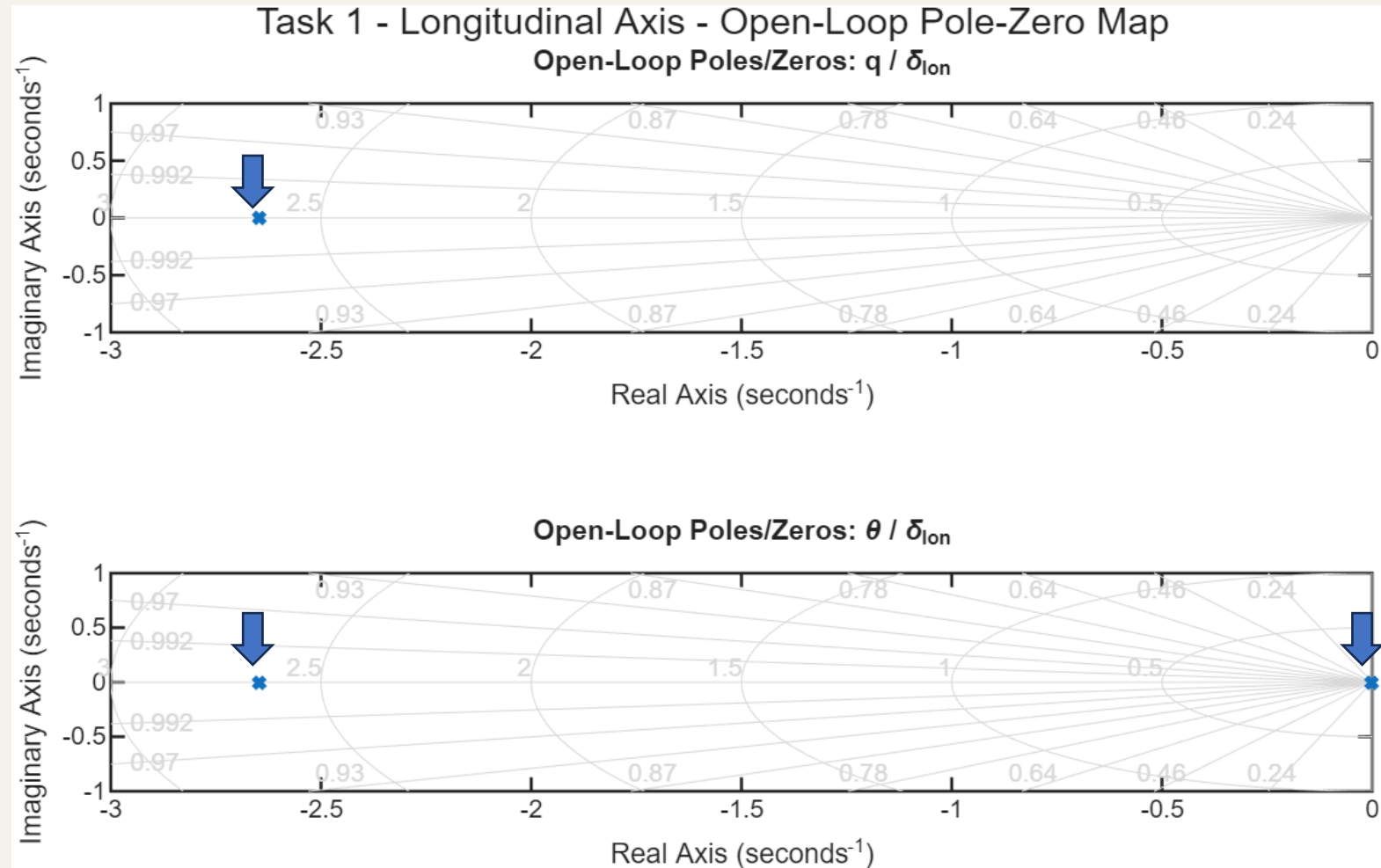
- Both p and ϕ diverge without feedback
- Stabilization, is the main challenge



Longitudinal axis — poles and zeros

Longitudinal

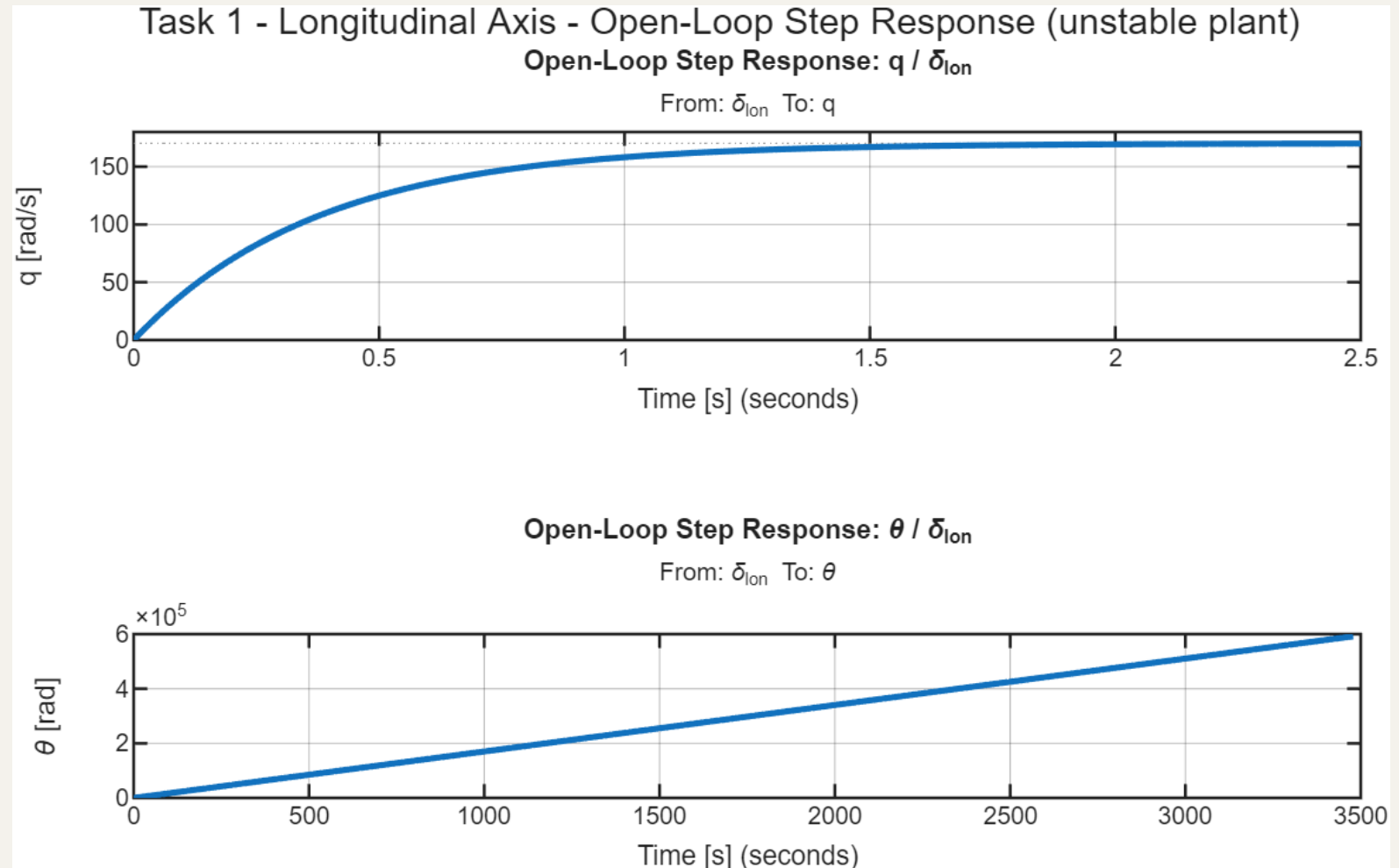
- Pitch mode at $\text{Re} = -2.6478$
the plant is **stable**
- Same fixed integrator on the angle channel



Longitudinal axis — step response

Longitudinal

- The rate settles, the angle ramps as expected



Performance Requirements

Of the reference second order system.

10 rad/s

minimum natural frequency ω_n

0.8

minimum damping ratio ξ

Required 0.9 then reduced to 0.8 due to limitations of performance.

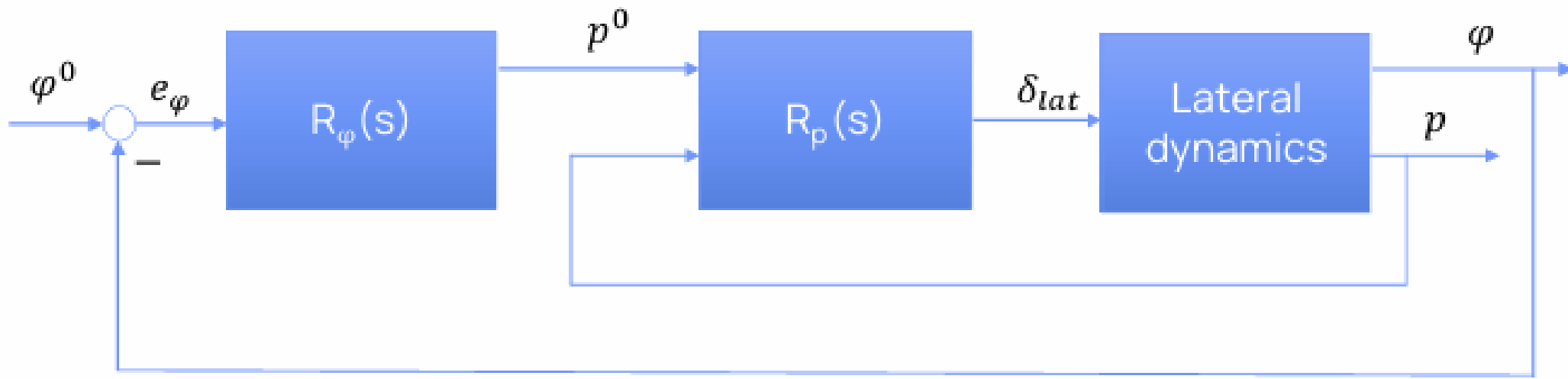
0.05

maximum command deflection $|\delta|$

It was not possible to maintain $\xi = 0.9$ with $\omega_n = 10$ rad/s because the required damping term would have forced the controller's derivative action to values incompatible with the physical actuator saturation constraint $|\delta| < 0.05$. Relaxing ξ was the necessary and sufficient mathematical condition to make the H_∞ synthesis problem feasible, ensuring the desired responsiveness without violating the control effort limits.

Architectural constraints imposed by the brief

- Rate controller R_p must be a filtered PID
- Angle controller R_ϕ must be a pure proportional gain
- Derivative filter time constant fixed at $T_p = 0.02$ s
- Cascade topology: angle outside, rate inside, with the PID being faster than the Proportional -> reason why we can have a cascade system.



The second-order prototype

Damping from overshoot

$$\xi_{eq} = \frac{-\ln(OS)}{\sqrt{\pi^2 + \ln^2(OS)}}$$

Equivalent 2nd-order value from the step-response overshoot (OS = fractional overshoot). $\xi_{eq} = 1$ when the response has no overshoot.

Frequency from rise time

$$\omega_{n,eq} = \omega_{n,ref} \frac{t_{r,ref}}{t_r}$$

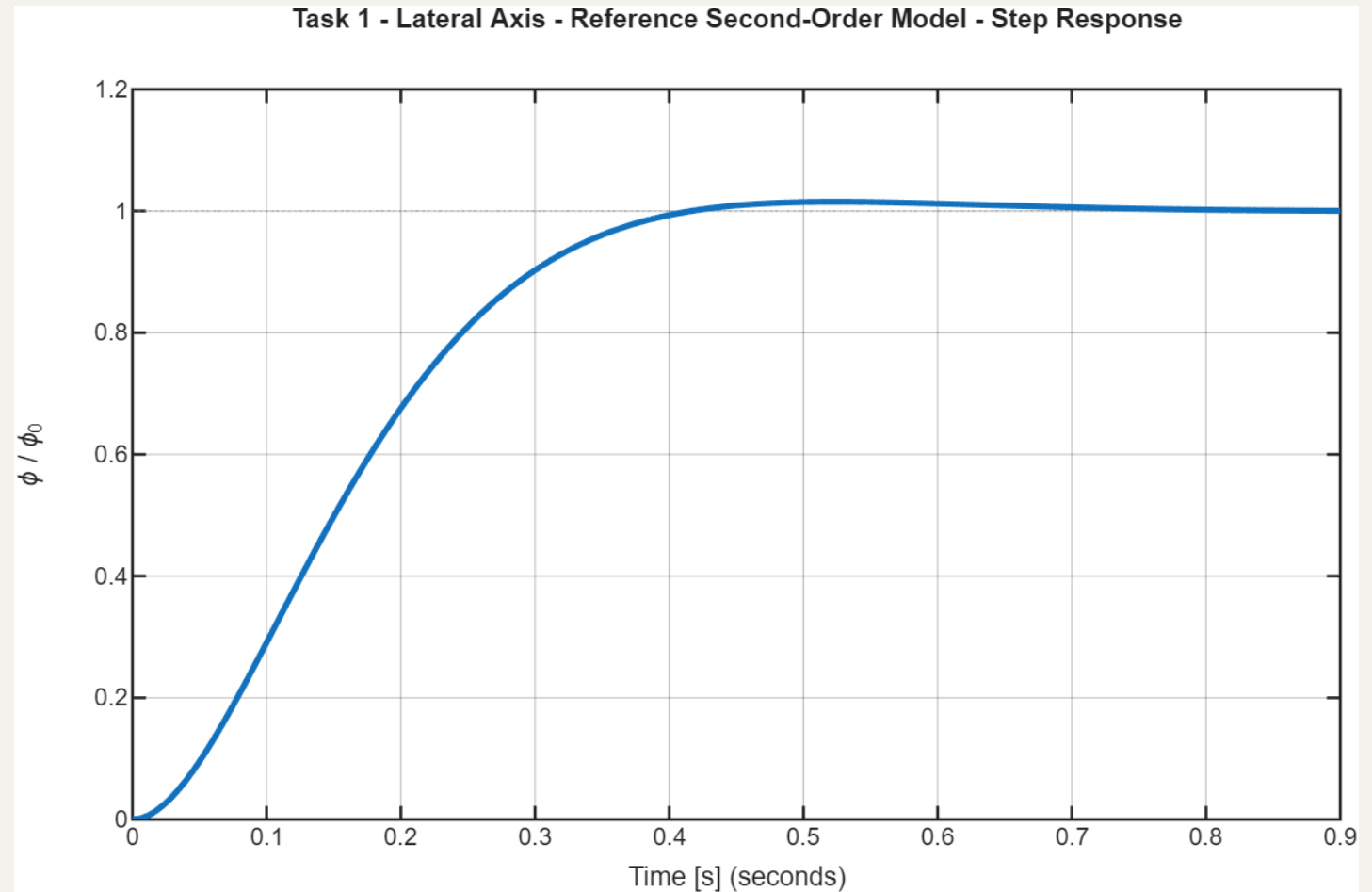
From the rise time, scaled against a 2nd-order reference ($\xi = 0.8$, $\omega_n = 10$): a faster response $\Rightarrow$ higher ω_n .

Response-based, not pole-based

Both metrics are read on the step response (equivalent 2nd-order), not on the dominant-pole locations.

Target second-order response

- F2 built with $\omega_n = 10$ rad/s and $\xi = 0.8$
- It is the shaping target, for the H_∞ weights.



Mixed sensitivity in one idea

Characteristic functions of the loop:

$$S(s) = \frac{1}{1 + L(s)} = 1 - T(s)$$

sensitivity

$$F(s) = \frac{L(s)}{1 + L(s)}$$

complementary sensitivity

Performance

$$\| W_p S \|_\infty < 1$$

W_p sets the allowed ceiling for $|S|$ (disturbance rejection, bandwidth).

Control effort

$$\| W_q K S \|_\infty < 1$$

W_q limits $|K \cdot S|$, i.e. actuator usage across frequency.

The weights: performance and control effort

W_p — performance

$$W_p = \frac{s/M + \omega_b}{s + \omega_b A}$$

A low-frequency ceiling: steady-state error and disturbance rejection ($A \uparrow \Rightarrow \text{error} \uparrow$).

M high-frequency ceiling: peak of $|S|$ and damping ($M \uparrow \Rightarrow \xi \downarrow$).

ω_b required bandwidth: $\omega_b \uparrow \Rightarrow \omega_n \uparrow$, but needs higher K and $|\delta|$ grows.

W_q — control effort (high-pass)

$$W_q = W_{HF} \frac{s + \omega_c W_{LF}/W_{HF}}{s + \omega_c}$$

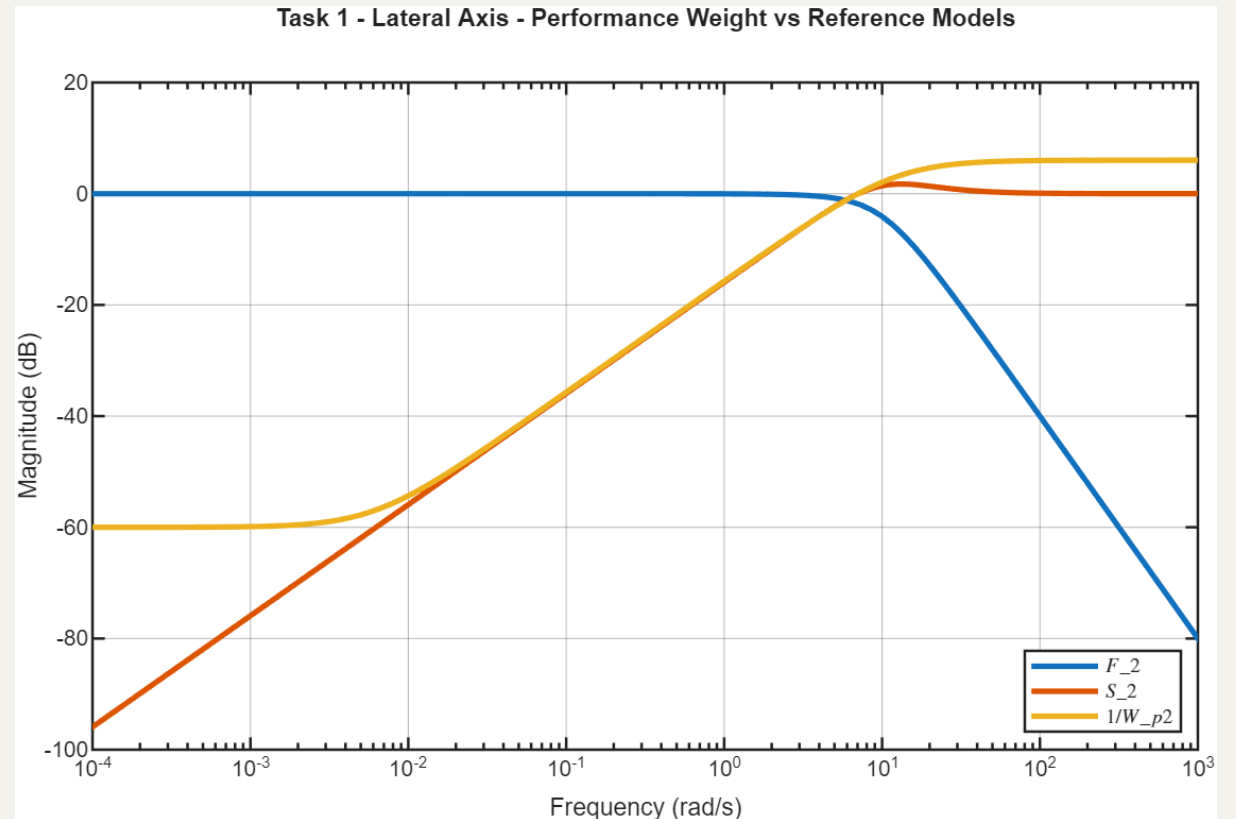
W_{HF} high-frequency gain: limits δ in fast maneuvers such as the doublet ($W_{HF} \uparrow \Rightarrow \omega_n \downarrow$).

W_{LF} low-frequency limit: contains δ in slow, steady commands.

ω_c sets the transition between slow and fast command.

Performance bound against the reference

- $1/|W_p|$ is the ceiling the synthesised $|S|$ must respect
- Overlaid on F2 and S2 to sanity-check the target



Weights Tuning

Lateral Performance W_p

- $\omega_b = 0.61 \cdot \omega_n$
- Higher demand to stabilize the RHP pole
- $M_{lat} = 2$;
- $A_{lat} = 1e-3$;

Lateral Performance W_p

- Corner at $\omega\tau = 20$ rad/s, the actuator roll-off
- Gain α scaled on a 20 deg maneuver and the peak deflection
- $\alpha_{lat} = \text{deg2rad}(20)/0.0715$;

Longitudinal Performance W_p

- $\omega_b = 0.59 \cdot \omega_n$
- Slightly relaxed, no pole to move
- $M_{lon} = 2$;
- $A_{lon} = 1e-3$;

Lateral Performance W_p

- Corner at $\omega\tau = 20$ rad/s, the actuator roll-off
- Gain α scaled on a 20 deg maneuver and the peak deflection
- $\alpha_{lon} = \text{deg2rad}(20)/0.0645$;

Lateral – step response against the reference

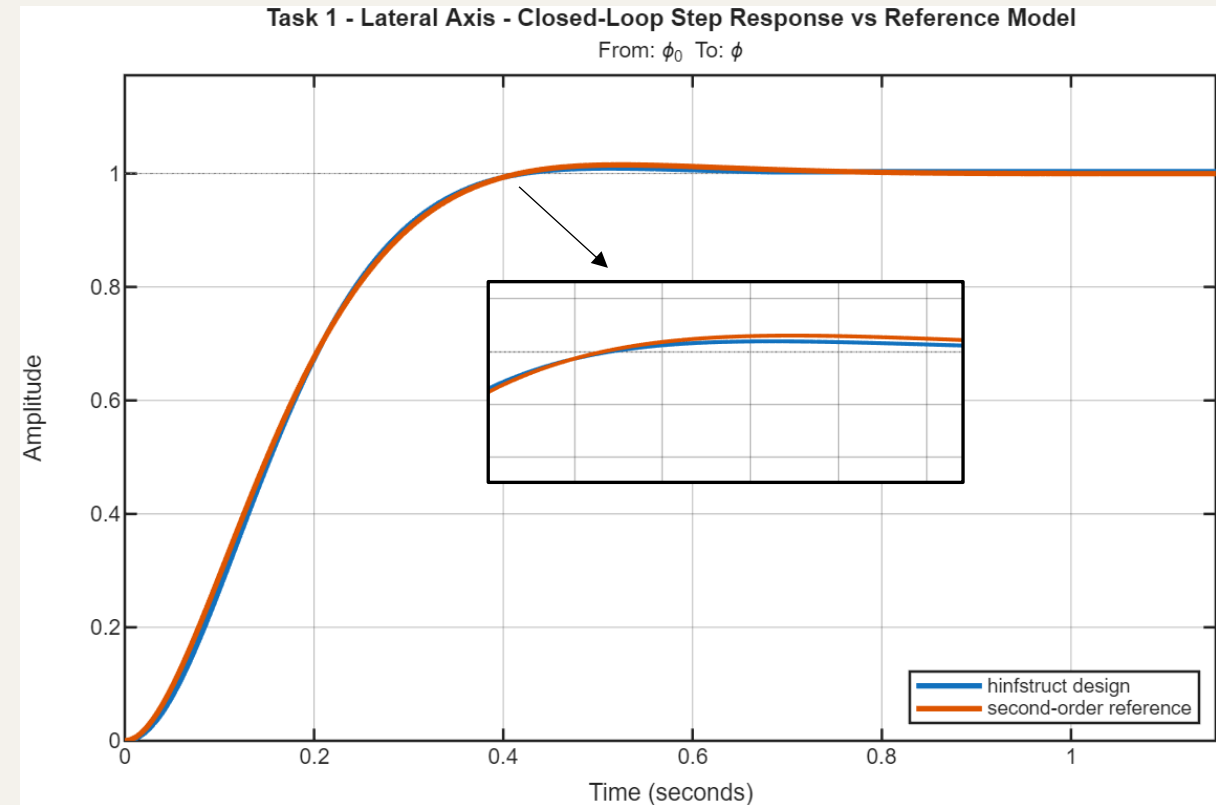
- Synthesized response overlaid on the second-order target
- Rise time and overshoot feed the equivalent metrics

StepInfo Closed Loop:

RiseTime: 0.2374
TransientTime: 0.3728
SettlingTime: 0.3728
SettlingMin: 0.9031
SettlingMax: 1.0098
Overshoot: 0.9782
Undershoot: 0
Peak: 1.0098
PeakTime: 0.5128

Stepinfo Second Order:

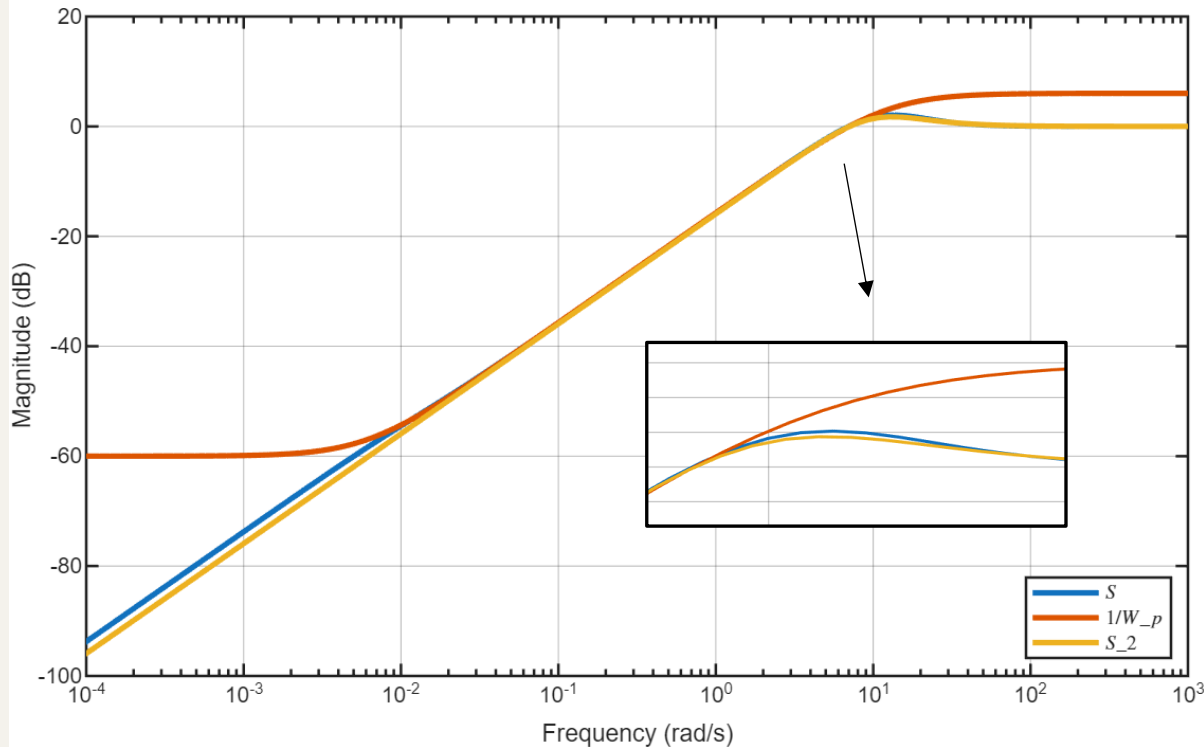
RiseTime: 0.2468
TransientTime: 0.3756
SettlingTime: 0.3756
SettlingMin: 0.9018
SettlingMax: 1.0152
Overshoot: 1.5165
Undershoot: 0
Peak: 1.0152
PeakTime: 0.5238



Lateral – Frequency Domain Response

Task 1 - Lateral Axis - Sensitivity Function S vs Design Bound

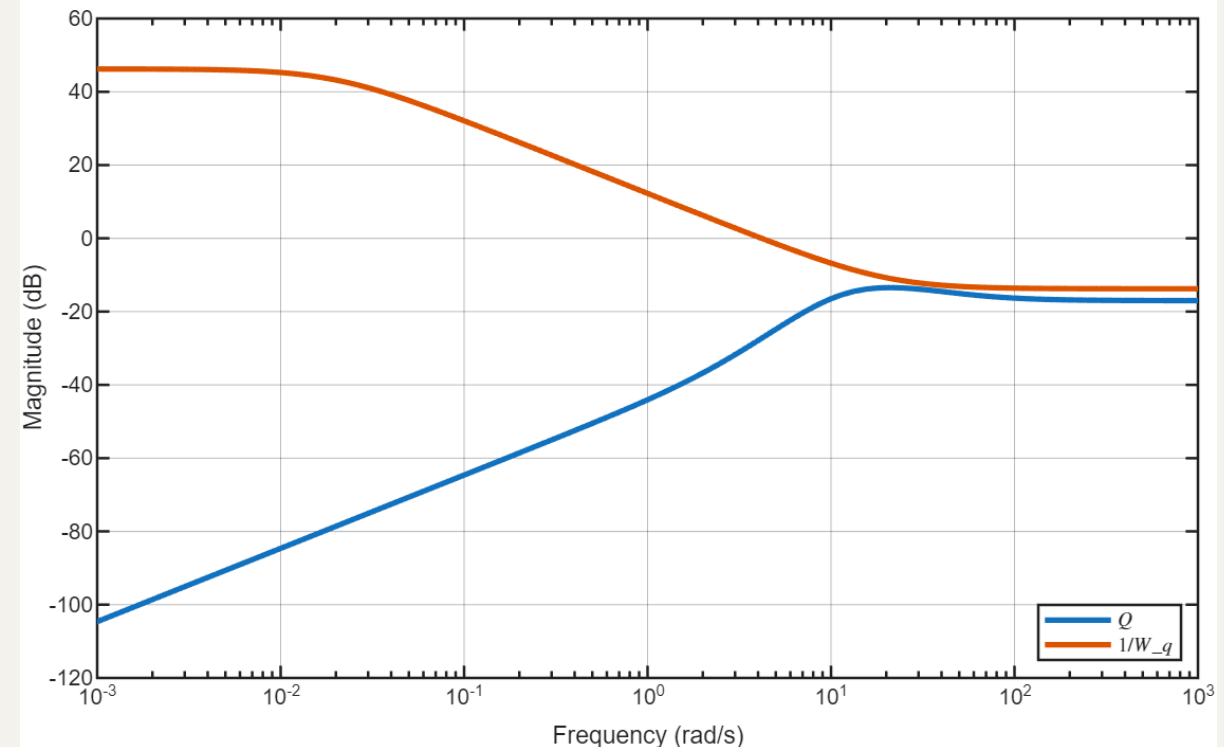
From: ϕ_0 To: e_ϕ



$|S|$ stays below the $1/|W_p|$ ceiling: performance and disturbance rejection are met.

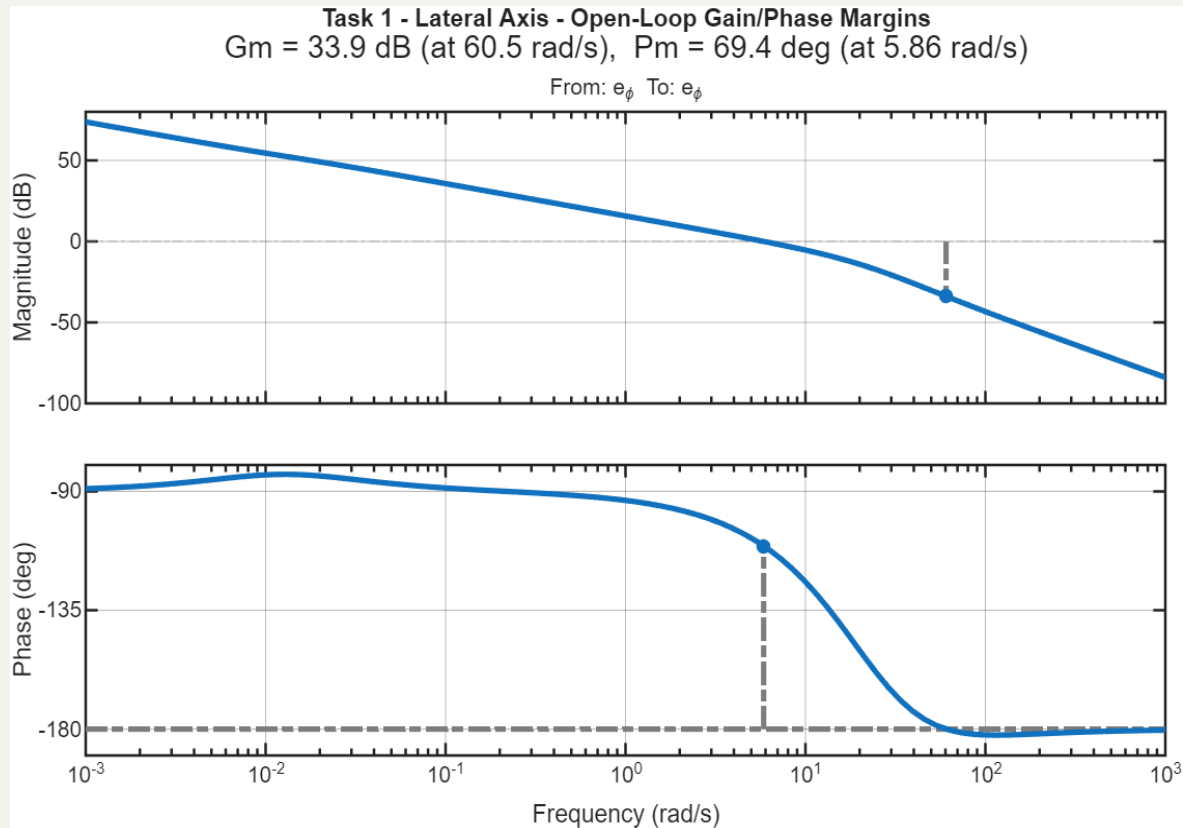
Task 1 - Lateral Axis - Control Sensitivity Q vs Design Bound

From: ϕ_0 To: δ_{lat}

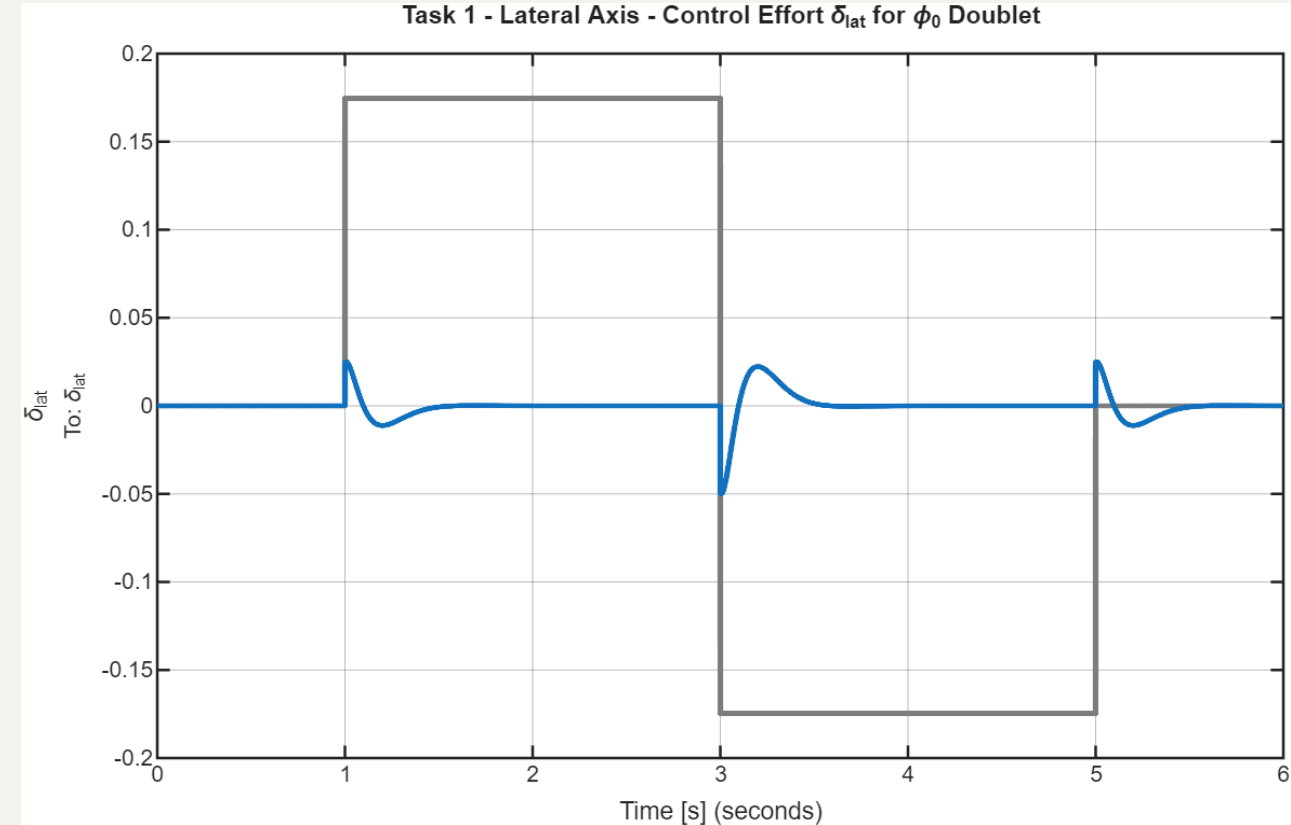


$|K \cdot S|$ stays below $1/|W_q|$: control effort is bounded across the band.

Lateral – Phase margin and Doublet Response



Positive gain and phase margins.



$\max|\delta_{lat}| = 0.04990$ (req ≤ 0.05)

"There is no such thing as enough control authority"

Lateral axis — requirement check

Quantity	Requirement	Obtained
γ	< 1	1.0204
ω_n equivalent	≥ 10 rad/s	10.3965 rad/s
ξ equivalent	≥ 0.9	0.8273
$\max \delta_{lat} $	≤ 0.05	0.04990

Longitudinal — step response against the reference

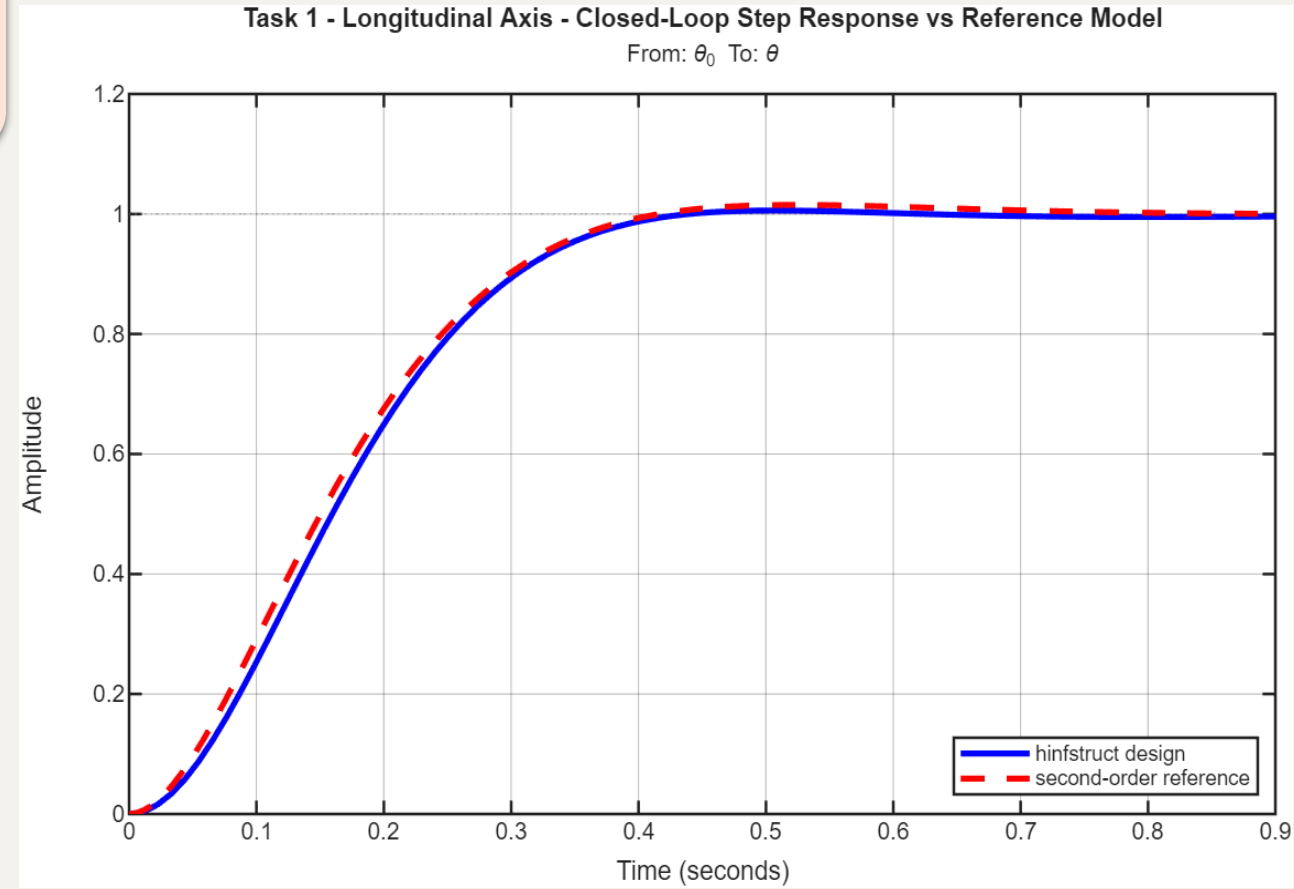
- Synthesized response overlaid on the second-order target
- Rise time and overshoot feed the equivalent metrics

Stepinfo Closed-loops

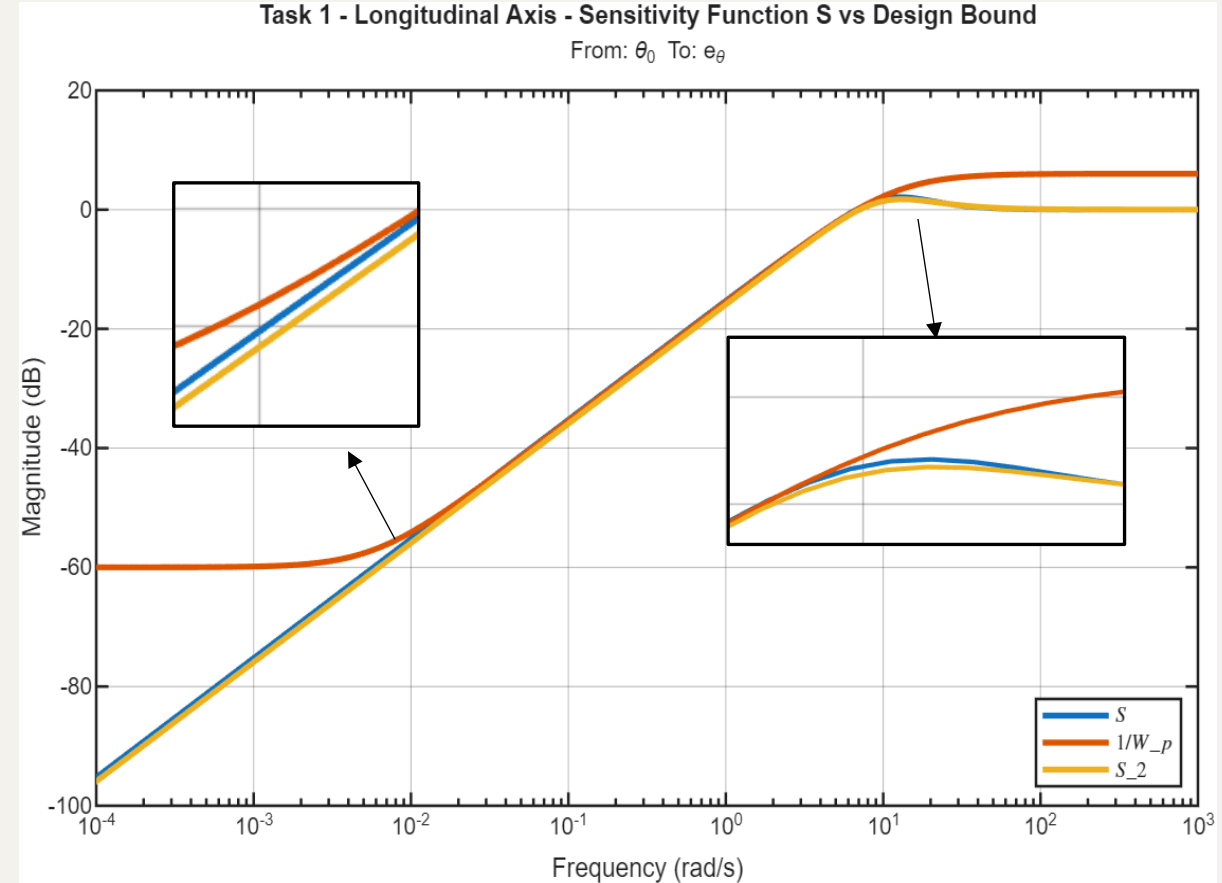
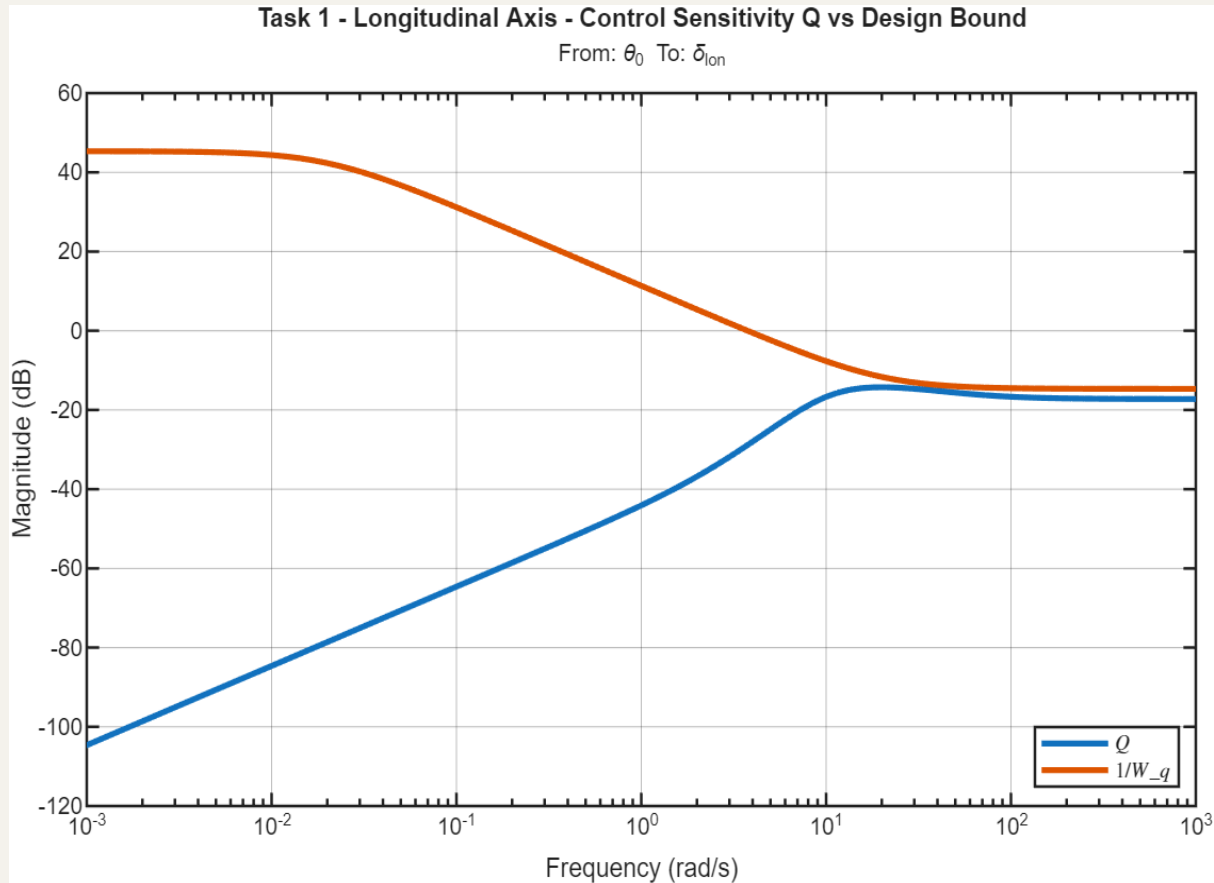
RiseTime: 0.2458
TransientTime: 0.3851
SettlingTime: 0.3851
SettlingMin: 0.9033
SettlingMax: 1.0058
Overshoot: 0.5809
Undershoot: 0
Peak: 1.0058
PeakTime: 0.5033

StepInfo Second Order:

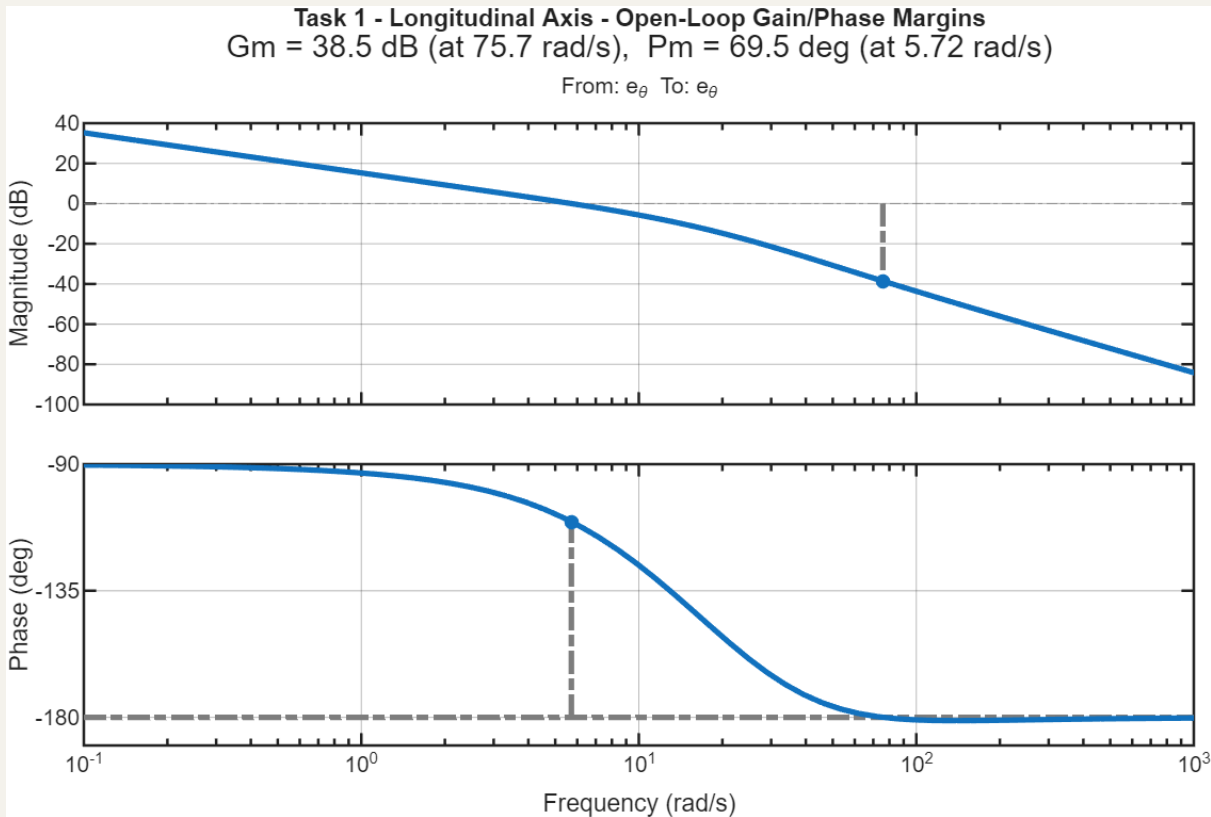
RiseTime: 0.2468
TransientTime: 0.3756
SettlingTime: 0.3756
SettlingMin: 0.9018
SettlingMax: 1.0152
Overshoot: 1.5165
Undershoot: 0
Peak: 1.0152
PeakTime: 0.5238



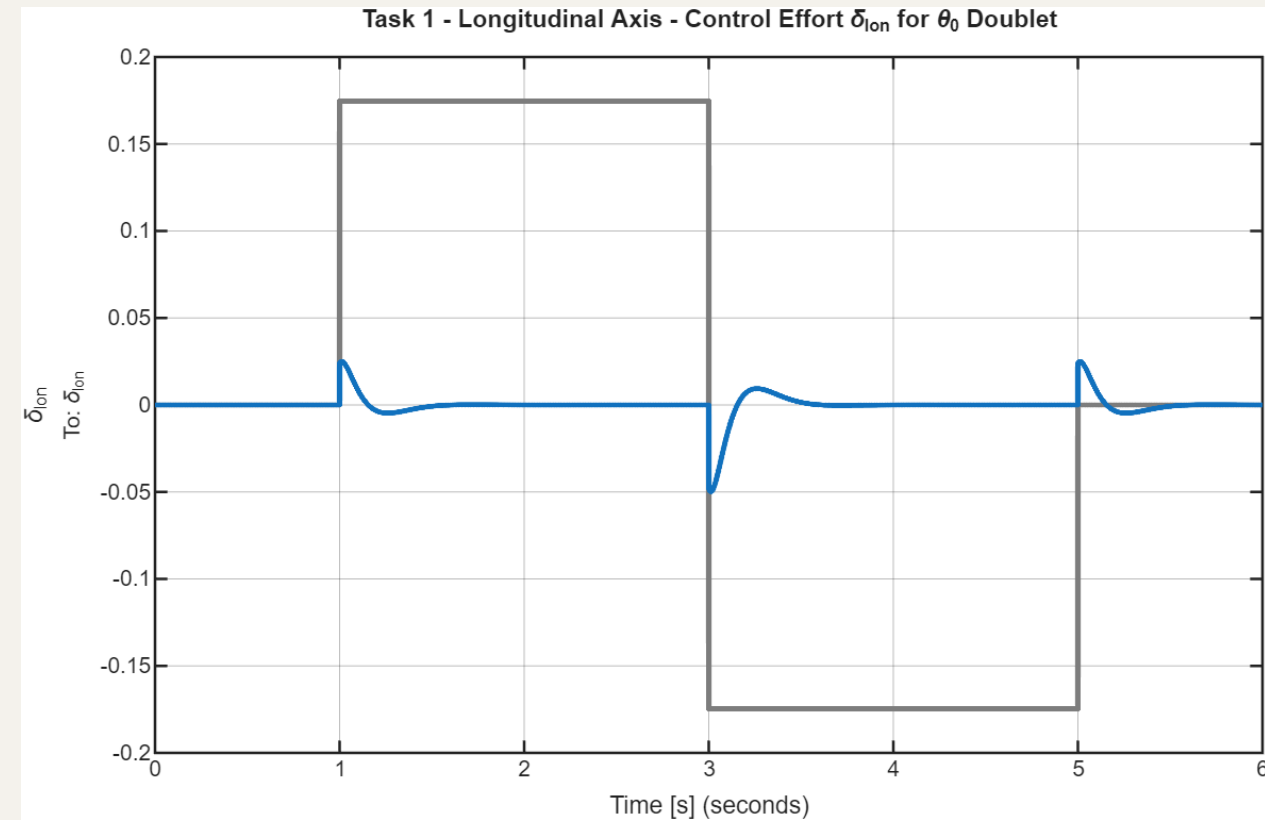
Longitudinal — sensitivity versus its bound



Longitudinal — margins and control effort



Positive gain and phase margins.



Longitudinal axis — requirement check

Quantity	Requirement	Obtained
γ	< 1	1.0193
ω_n equivalent	≥ 10 rad/s	10.0414
ξ equivalent	≥ 0.9	0.8536
$\max \delta_{lon} $	≤ 0.05	0.04973

TASK 1- SUMMARY

Conclusions

Longitudinal axis

- Plant already stable: control on performance only.
- $\gamma = 1.0193 \approx 1$: valid mixed-sensitivity synthesis.
- ξ (overshoot) ≈ 0.8536 and $|\delta| \leq 0.05$ satisfied.

Lateral axis

- Unstable plant: roll pole moved to LHP.
- Performance and effort within limits; $\gamma \approx 1$.
- Same P + PID architecture, retuned weights.

All requirements met in the response-based formulation — damping relaxed to 0.8 for the effort constraint; dominant-pole damping documented.

02

Uncertain modelling

Task 2 — from standard deviations to a deterministic family

From a density to a bound

- Identification returns a σ , not an interval
- Worst-case analysis needs a hard, finite set
- Three σ covers about 99.7% of a Gaussian
- The tail beyond it is deliberately excluded

Why parametric, not unstructured

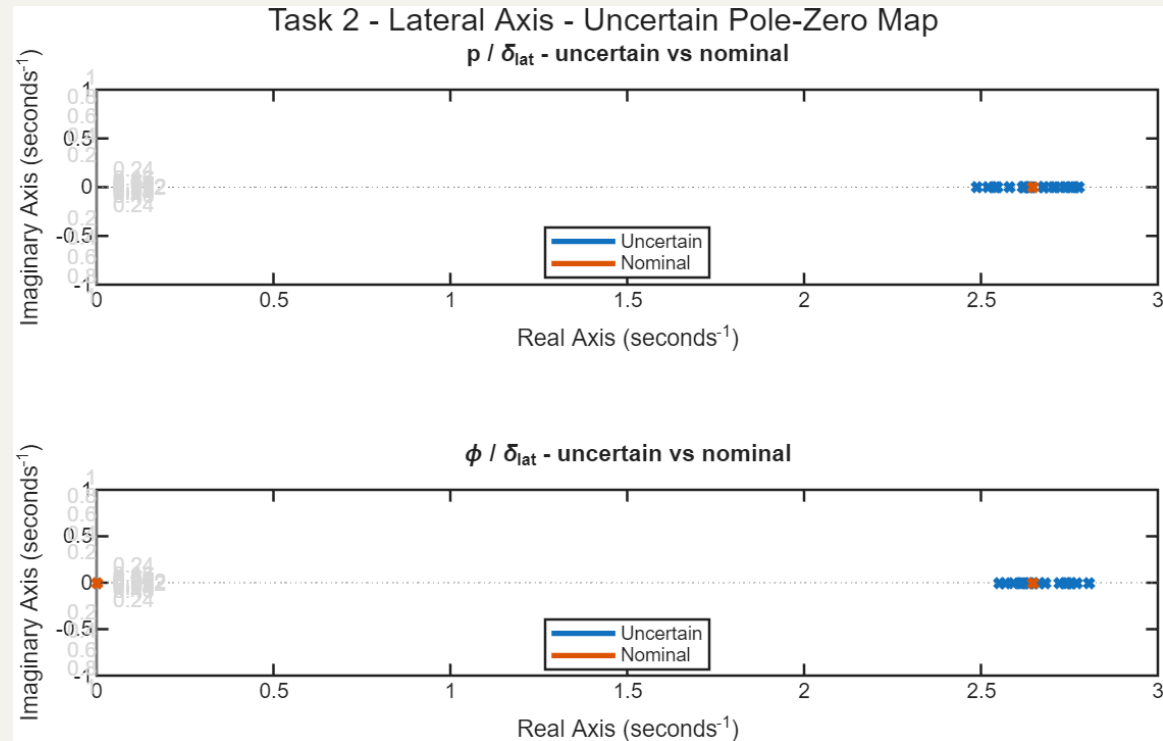
SUMMARY

- The uncertainty has a known physical origin
- Each derivative is one real, independent parameter
- An unstructured bound would cover impossible plants
- Keeping the structure keeps the analysis tight

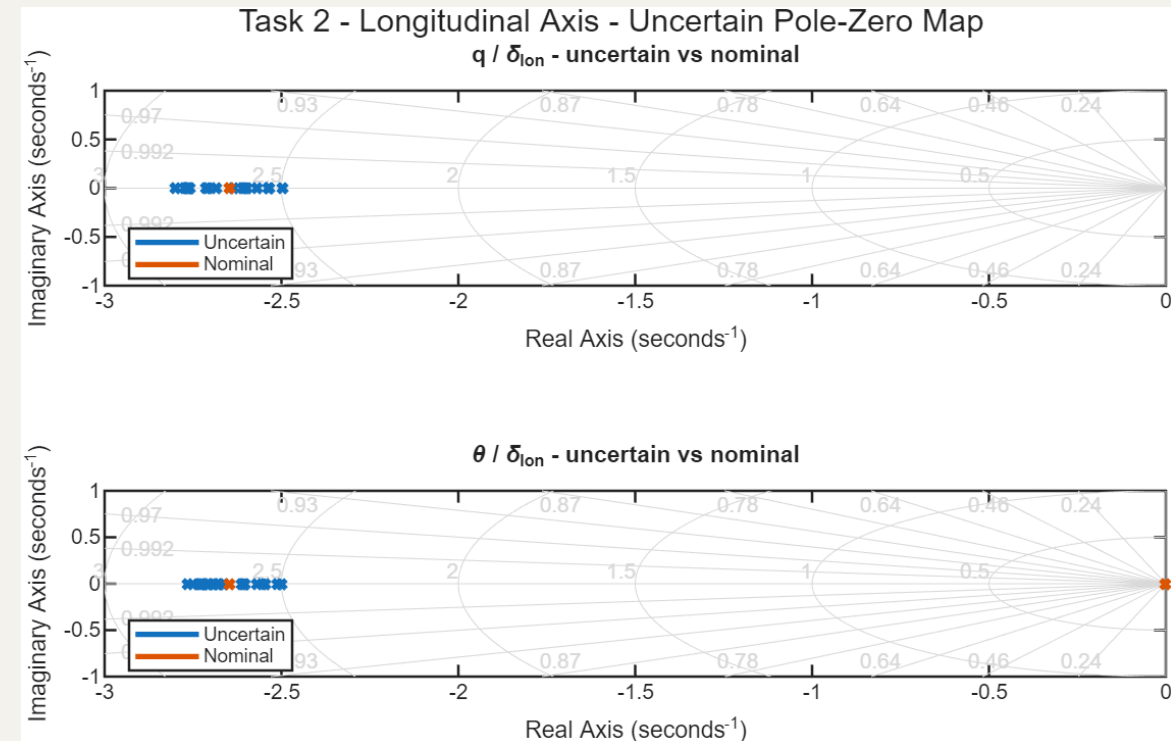
Building the uncertain plants

- Each derivative declared with ureal at three times its σ
- Five uncertain parameters per axis enter A and B
- The kinematic row carries no uncertainty at all
- `uss` assembles the family, `NominalValue` gives the centre

Lateral – Longitudinal – uncertain pole-zero map

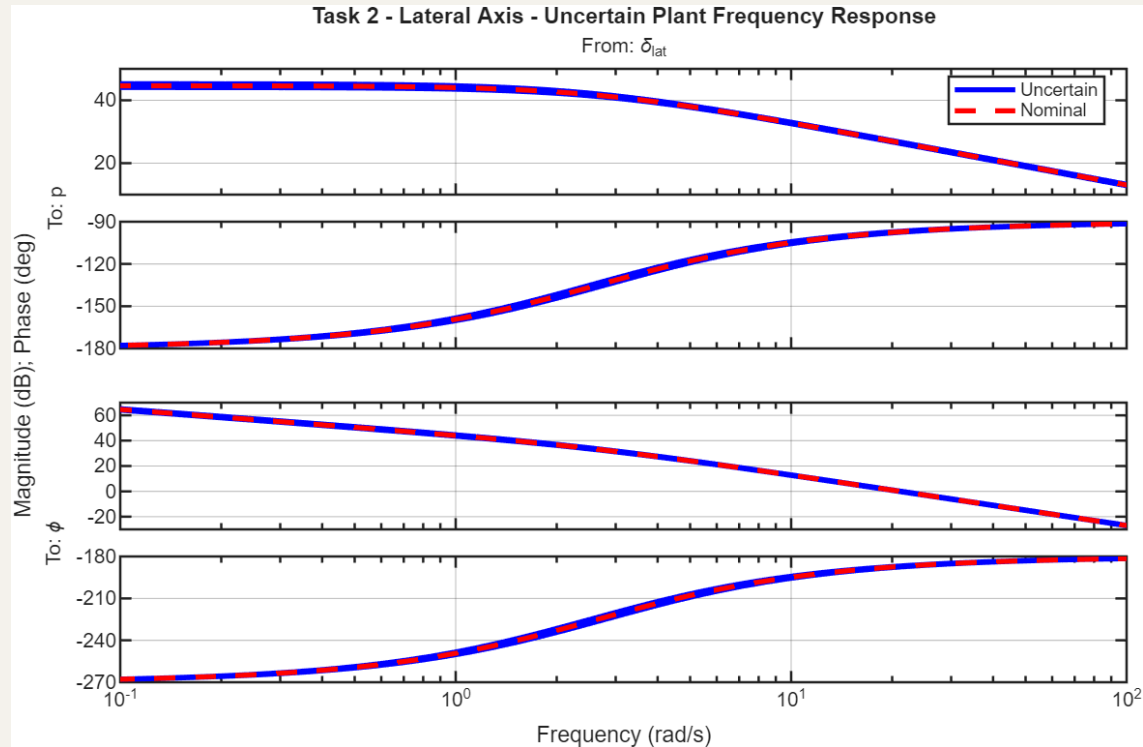


- The roll-pole cloud stays in the right half-plane
- The unstable pole count is invariant over the family

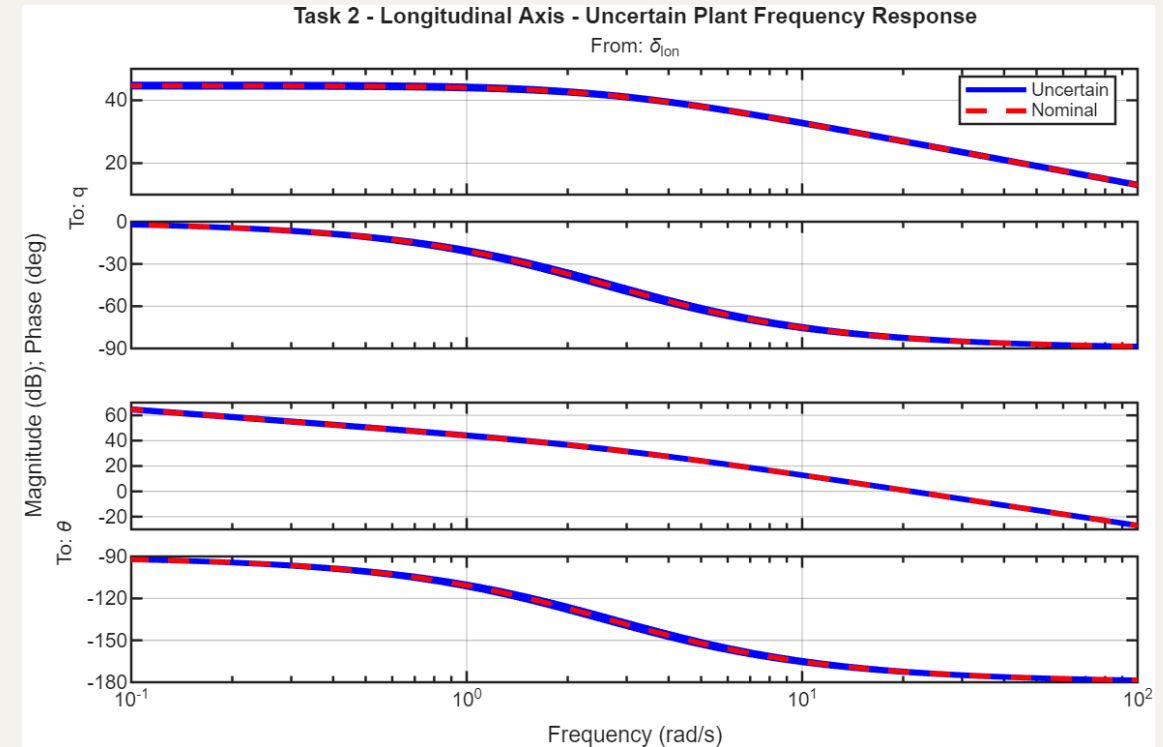


- The pitch cloud stays in the left half-plane
- No member of the family crosses the imaginary axis

Lateral-Longitudinal — uncertain frequency response



- A narrow beam around the nominal response
- The attitude channels depend mainly on L_p and L_δ



- Same narrow dispersion on the pitch channels
- The corner frequency moves with M_q alone

Why the beam is so narrow

- The most uncertain derivative lives in the velocity row
- That row is unobservable from the attitude outputs
- Only the rate and control derivatives reach the loop
- Effective uncertainty is therefore smaller than declared

03

Robustness analysis

Task 3 — M - Δ form, small gain and structured singular value

Robust Stability: The Core Question and Normalization

The core question

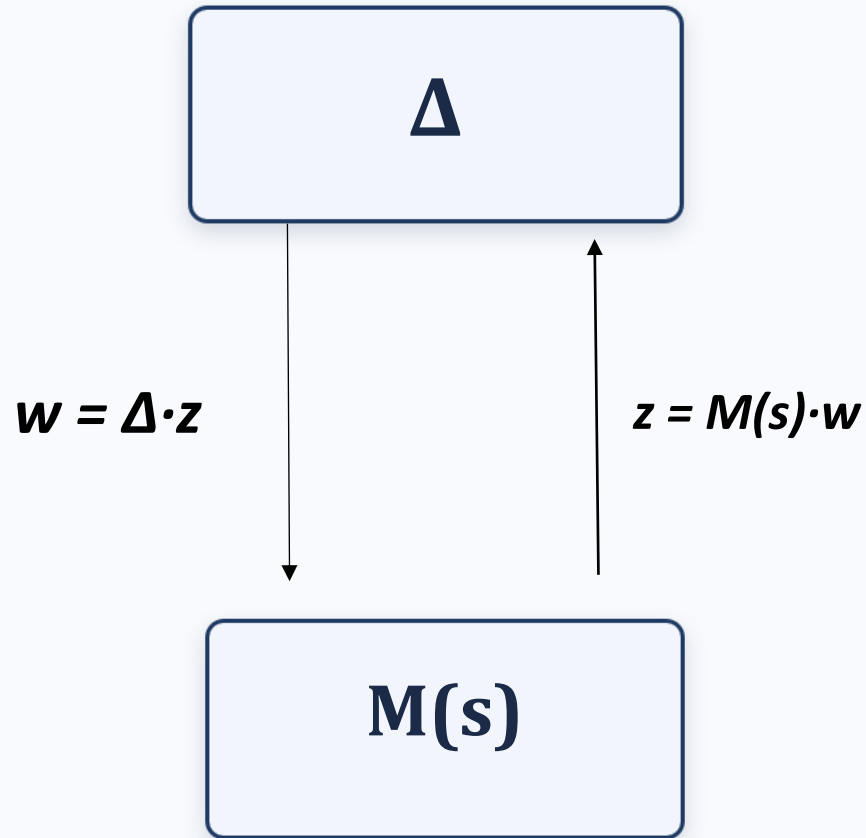
- The Task 1 controllers are frozen; no gains are retuned.
- The analysis determines if the dynamic loop remains stable for every possible parameter combination within the $\pm 3\sigma$ uncertain family, rather than just the nominal plant.

Normalization

$$\mathbf{p} = \mathbf{p}_0 + 3\boldsymbol{\sigma} \cdot \boldsymbol{\delta}, \quad \boldsymbol{\delta} \in [-1, 1]$$

- The 3σ scaling is absorbed into the known system, ensuring the uncertainty block Δ remains a normalized ($\|\Delta\| \leq 1$), block-diagonal matrix.
- This normalization sets the small-gain robust stability threshold to exactly 1.

M- Δ framework and implementation



- **M** is a known dynamical LTI system with its own state-space realization, not just a constant matrix.
 - Δ is the normalized uncertainty ($\|\Delta\| \leq 1$), block-diagonal, one entry per uncertain parameter.
-
- The *lftdata* function automatically splits the uncertain closed-loop model into the **M** and Δ matrices.
 - We isolate **M_11**, the specific submatrix of **M** that directly interfaces with the structured parametric uncertainties.

Two Approaches

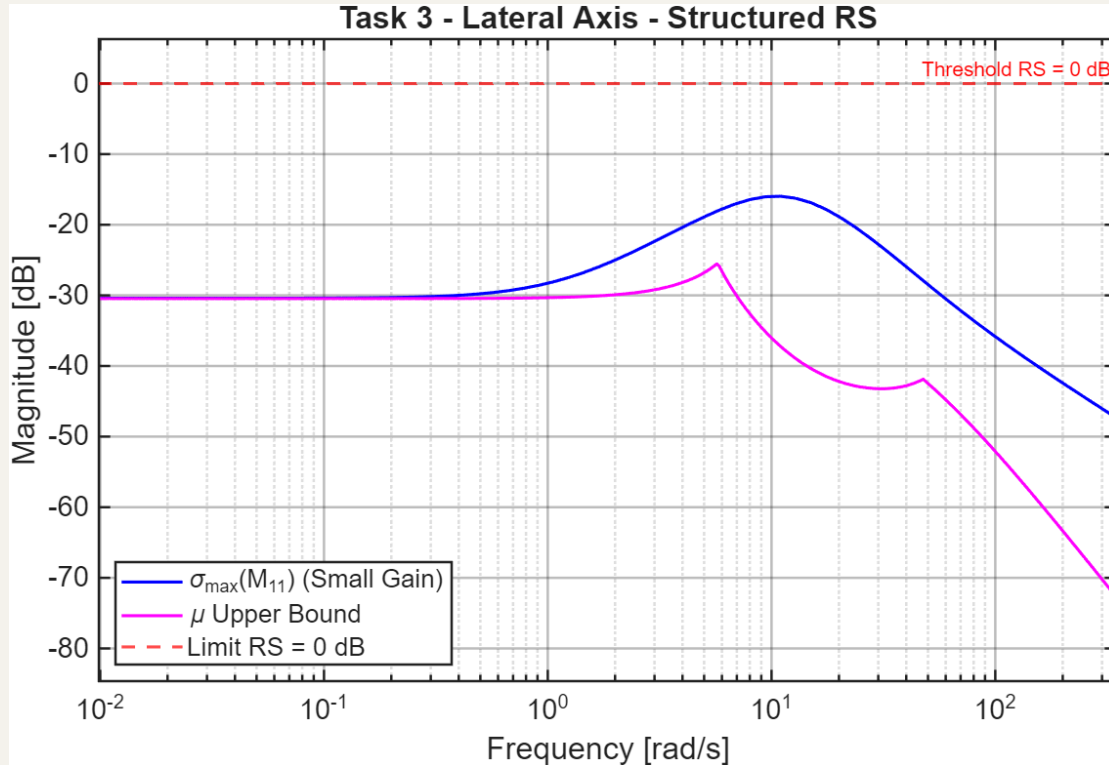
Small gain

- Treats Δ as complex and full
- Very conservative
- Overestimates the uncertainties
- Risk of overdesigning the controller
- H_∞ norm of M_{11} strictly less than 0 dB

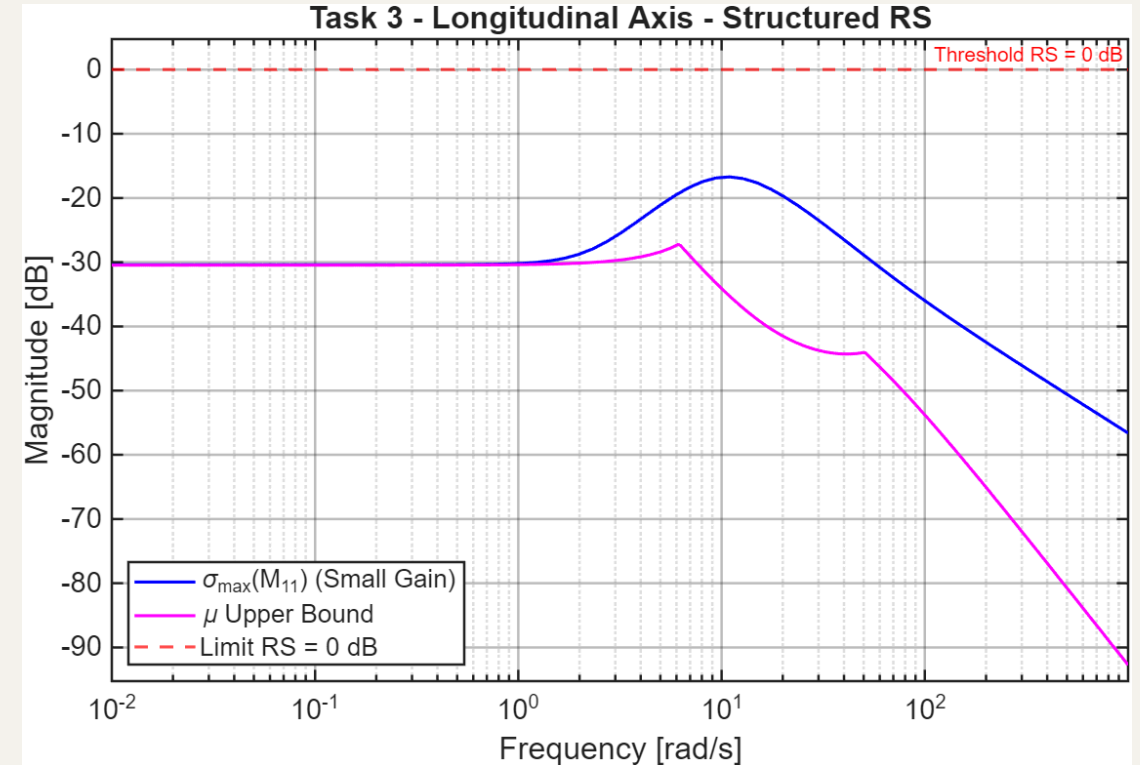
Structured μ

- Restricts Δ to real, diagonal blocks
- Worst case over the admissible set only
- More precise
- Upper Bound strictly less than 0 dB

Results — structured robust stability



- Small-gain curve and μ upper bound, in decibels
- Robust stability holds while both stay below 0 dB



- The gap between the curves measures the conservatism
- SSV has two peaks because more precise and can distinguish the effects of the controllers

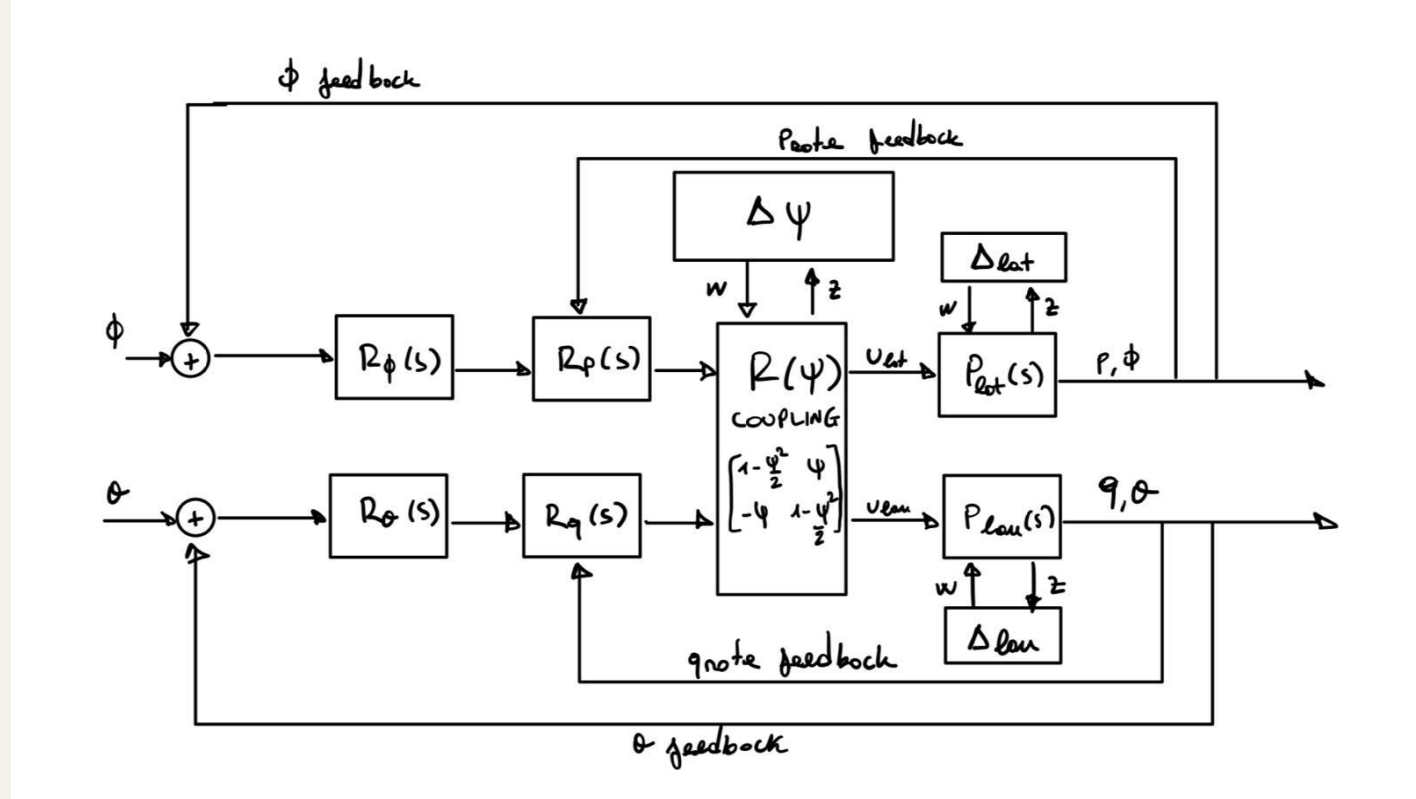
04

Robustness to coupling

Task 4 — uncertain coupling between the two axes

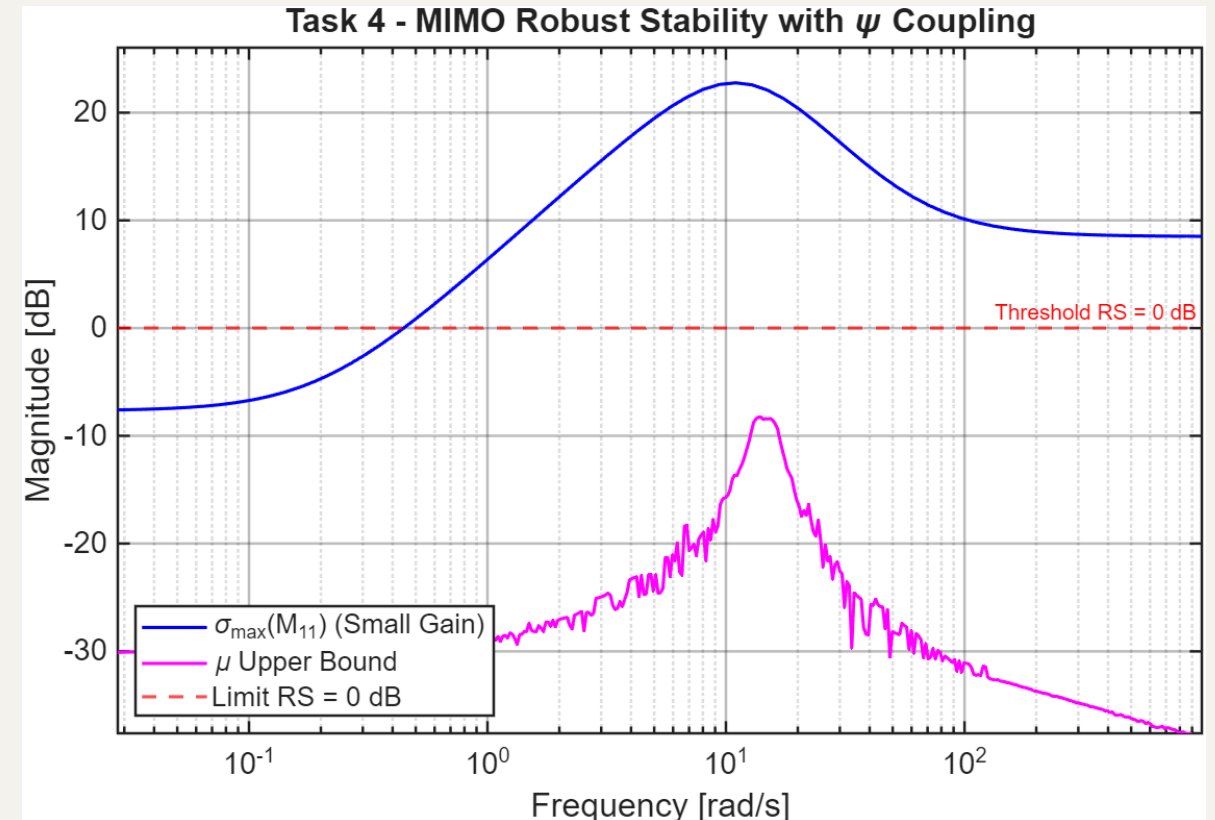
The coupling model

- Define longitudinal-lateral cross-coupling matrix, function of the uncertain angle ψ (range between -15° and $+15^\circ$)
- Apply Taylor series expansion : $\sin \psi = \psi$, $\cos \psi = 1 - \frac{\psi^2}{2}$
- Complete the MIMO closed-loop system, connecting: two plants, uncertain coupling matrix, and the four controllers



MIMO robust stability with coupling

- Small-gain curve and μ against the 0 dB threshold
- Small Gain's peak above 0 dB (no guaranteed robust stability) --> overestimation
- Structured Singular Value's peak below 0 dB (guaranteed robust stability)
- The sharp peak caused by the coupling matrix (both axes crossing at the same frequency)



05

Control system validation

Task 5 — Monte Carlo analysis of the complete loop

Worst case versus population

Deterministic analysis

- Answers yes or no on the whole set
- Says nothing about how likely failure is
- Becomes impractical with many parameters

Probabilistic analysis

- Estimates the fraction of compliant units
- Uses the real densities, not a hard box
- Cost independent of the parameter count

Monte Carlo Analysis

Uncertainties:

- Five aerodynamic derivatives on the lateral axis
- Five more on the longitudinal axis
- Gaussian densities centered on the nominal values
- Coupling angle ψ drawn uniformly over its range
- 11 sources of uncertainty in total
- M- Δ approach is impractical for these many sources

Goal:

- To estimate the fraction of the drone population that meets the requirements of the Task 1

The estimator and the Chernoff bound

- **Purpose:** To determine *a priori* the minimum number of iterations (N) needed to estimate the system's reliability via Monte Carlo analysis.
- **Statistical Advantage:** The error bound is completely independent of the high number of uncertain sources (11 parameters in total) and decays exponentially as the number of samples increases.
- **Rigorous Sizing:** Enforcing an accuracy of **2%** ($\epsilon = 0.02$) with a **95%** confidence level ($\delta = 0.05$), the theorem requires a minimum of **4612** simulations.
- **Project Application:** A total of **4812** iterations were used, including a conservative margin to guarantee the complete mathematical robustness of the result.

$$N \geq \frac{\ln(2/\delta)}{2\epsilon^2}$$

The resulting experiment

4612

minimum samples from the bound

4812

samples actually simulated

$\pm 2\%$

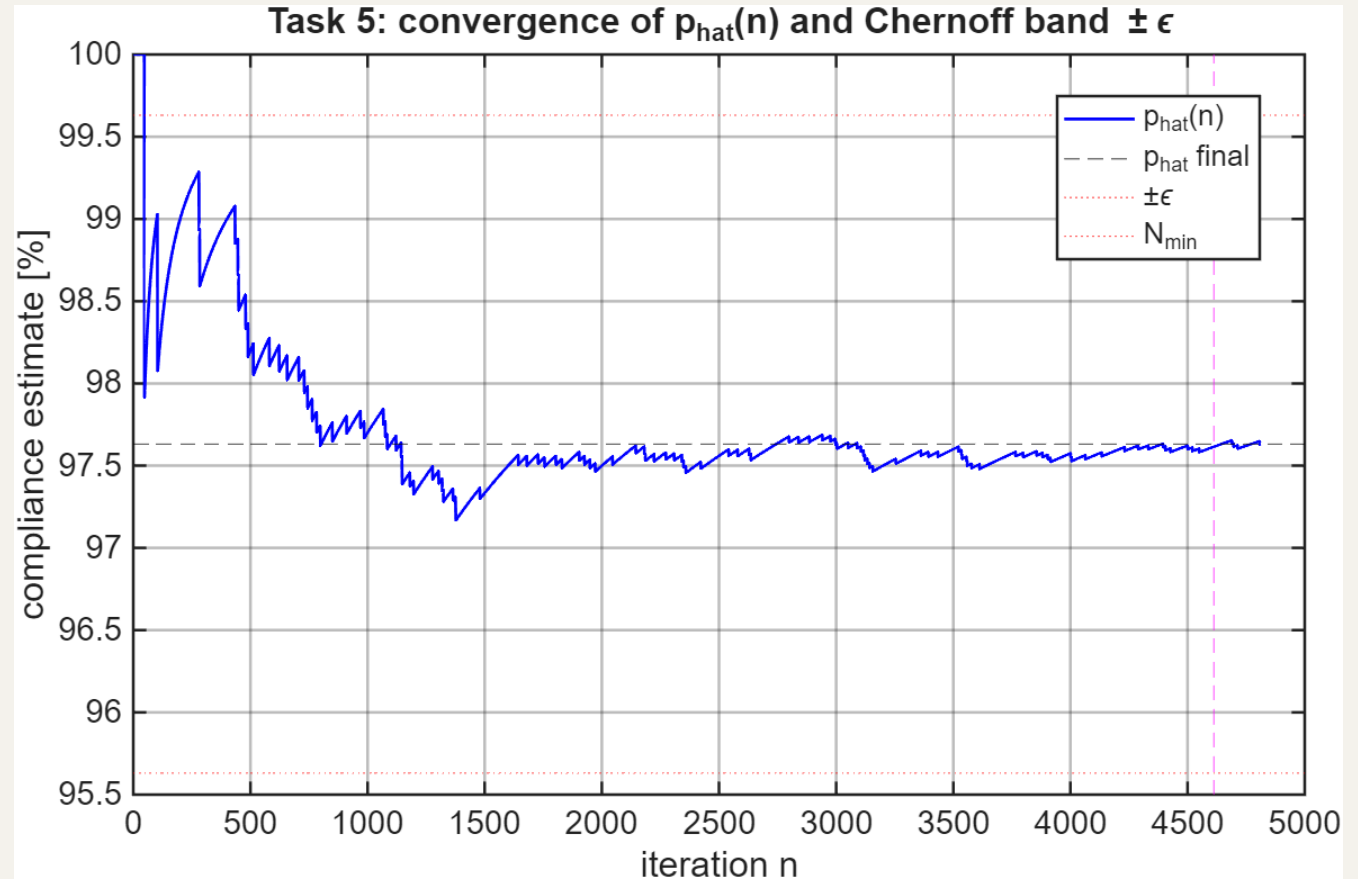
accuracy at 95% confidence

What each iteration does

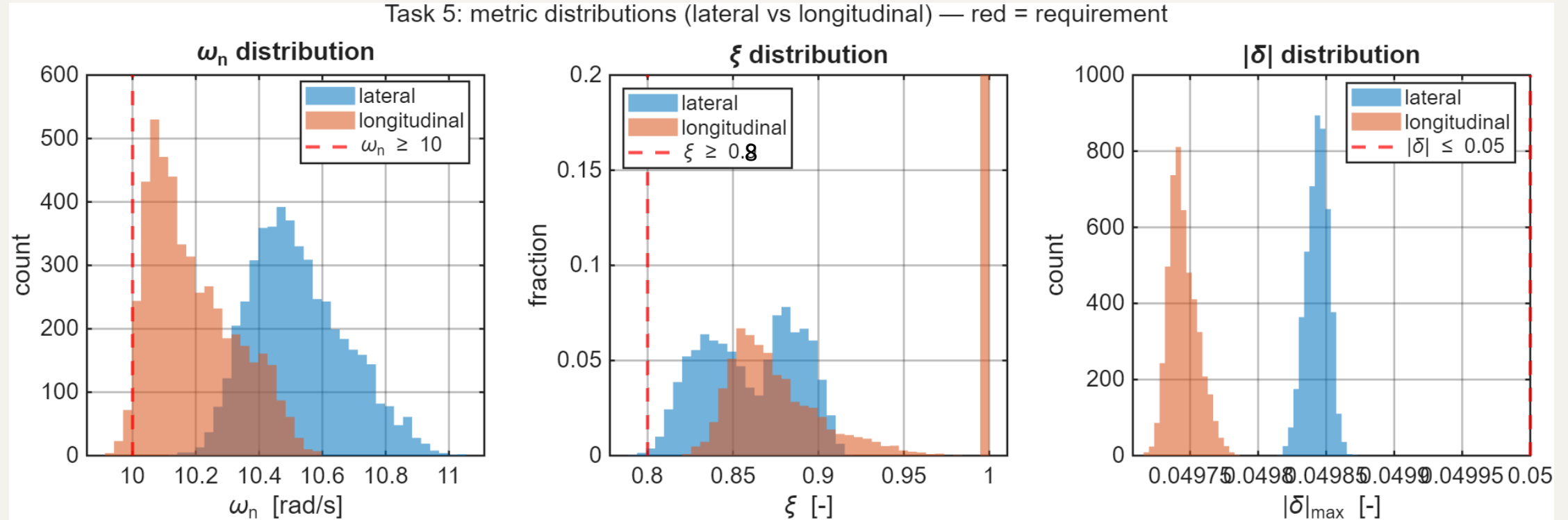
- Draw the ten derivatives from their densities
- Draw ψ uniformly between -15 and +15 degrees
- Rebuild both plants and the coupling rotation
- Close the loop with the frozen Task 1 controllers

Convergence of the compliance estimate

- Running estimate plotted against the Chernoff band
- Staying inside the band validates the chosen N
- $P_{\text{hat}} \approx 97.6\%$ of the population meets the task 1 requirements with $\pm 2\%$ accuracy at 95% confidence.



Distribution of the requirement metrics



- ω_n , ξ and $|\delta|$ for both axes, requirement in red
- Reveals which requirement is the binding one, in our case ω_n _lon
- As we can see $|\delta|$ is basically deterministic with no statistical dependency

Monte Carlo — code walkthrough

- A fixed seed makes the whole campaign reproducible
- randn and rand draw the parameters and the angle: Gaussian distribution (randn) is applied to the 10 aerodynamic parameters, while a Uniform distribution (rand) covers the cross-axis coupling in $\psi[-15 \text{ deg}, 15 \text{ deg}]$.
- connect rebuilds the complete loop inside the iteration
- Cumulative means produce the convergence curve

Thank you for your attention

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